

Math 300: Midterm 1 Review

June 27, 2023

1 Definitions you need to know

1. Even and Odd Numbers, and parity
2. Divisor and multiple, prime numbers
3. Division Algorithm
4. Greatest common divisor and least common multiple
5. Congruence modulo n
6. Rational and Irrational numbers
7. Basic properties of inequalities

2 Set Theory

1. Writing sets by listing or using set-builder notation
2. Elements, Subsets, Power Set, Indexed sets
3. Intersection, Union, Set Difference, Complement, Cartesian Product
4. Venn Diagrams

3 Logic

1. Statements
2. Truth Tables
3. And, Or, Not, Conditional, Biconditional
4. Logical Equivalence
5. Quantifiers (Existential, Universal)
6. Negation, both of statements and quantifiers, DeMorgan's Laws.

4 Writing Proofs

4.1 Proof Techniques

1. Direct Proof, proof by cases
2. Disproving using counterexamples
3. Proof by Contrapositive
4. Proof by Contradiction

4.2 Proof Structure

1. Deciding which technique to use, stating the technique and the assumptions considered
2. Using definitions, lemmas or life facts
3. Writing style, organization and clarity
4. Logical deduction

5 Midterm Format

1. 8 questions for 150 points
2. 1 bonus question worth 15 (extra) points
3. Several proofs where you will **not** be told which technique to use
4. The proofs will mostly be ones you write yourself, but there may also be short answer/explanation type questions.